NIYUKTI

To bridge the gap between student preparation and placement opportunities through AI-powered, data-driven placement management.

Team Atom

- Aayush Rathore
- Abhyansh Sahu
- Ansh Vardhan Dixit
- Nitin
- Priyanshu Shrivastava
- Rajnish Kumar
- Praphull Parnav
- Manshu Devi
- Shiwani Jha
- Shri Sharma
- Shruti Pandey
- Vani Singh

Problem statement

- **T&P teams** spend significant time managing student data, checking company-wise eligibility, processing applications, conducting assessments, shortlisting candidates and maintaining placement records.
- **Students** use separate platforms for resumes, coding, aptitude, technical preparation and interviews, making their preparation and placement journey fragmented.
- **Assessment, interview and student-performance data** remains scattered, making it difficult to get a complete view of a student's placement readiness.
- **Manual shortlisting and analysis** become difficult when the number of students and placement drives increases.
- T&P teams have limited **real-time insights into student skill gaps, performance and placement progress**.

Proposed Solution

Niyukti is an AI-powered placement management platform that connects student preparation with the complete T&P placement process.

- **Centralized Student Profile** — academics, skills, resume, projects, assessments and placement history.
- **T&P Placement Management** — companies, placement drives, eligibility, applications, assessments, interviews and selection.
- **AI-Based JD Analysis & Matching** — analyzes company requirements and matches them with student profiles and skills.
- **AI-Assisted Shortlisting** — helps T&P identify relevant candidates based on eligibility and performance.
- **Assessment & Performance Analysis** — aptitude, technical and coding assessments with centralized results.
- **Skill Gap & Personalized Preparation** — identifies areas for improvement and recommends relevant preparation.
- **AI Mock Interviews** — technical and HR interview practice with structured feedback.
- **T&P Analytics** — placement funnel, student performance, skill gaps, drive-wise insights and reports.

What Niyukti Offers the T&P Department

Student Progress Monitoring – Track students' skills, projects, certifications, learning progress, and internship readiness through a centralized dashboard.

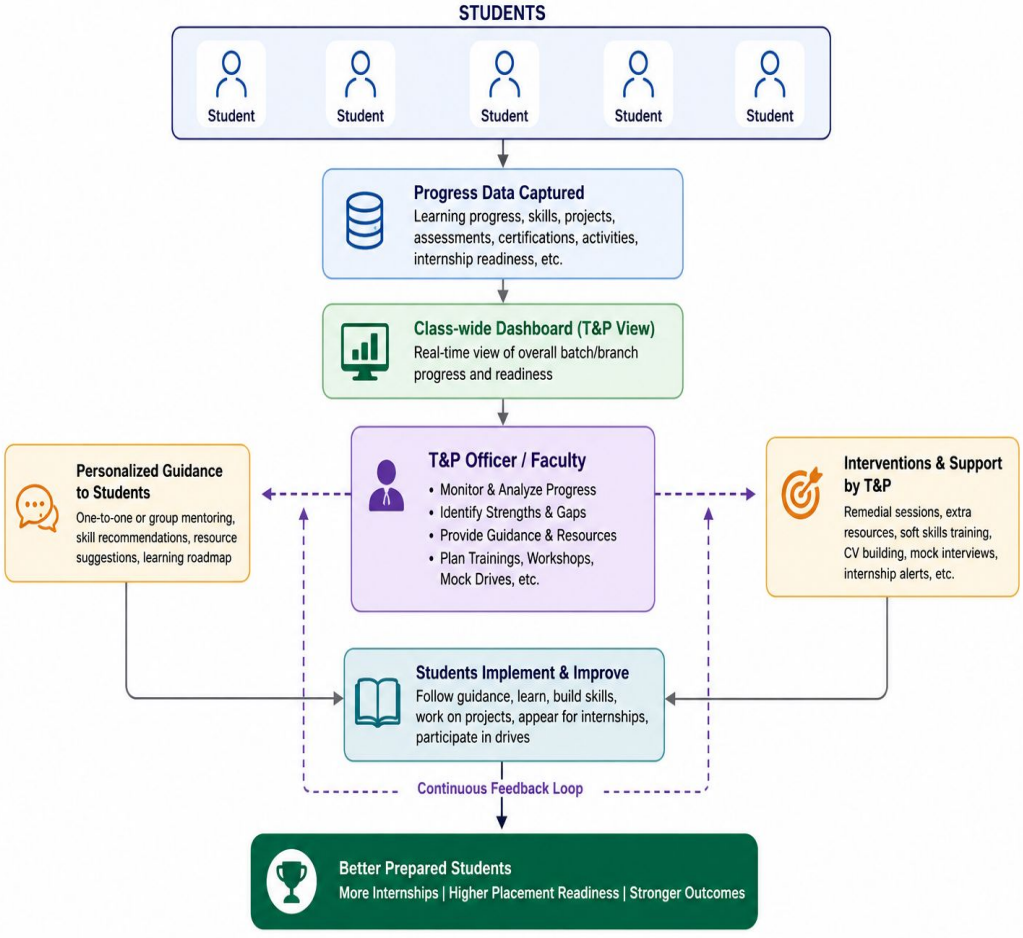
Company & Drive Management – Manage companies, internship opportunities, placement drives, eligibility criteria, roles, packages, and deadlines.

AI-Assisted Shortlisting – Automatically identify and shortlist eligible candidates based on skills, progress, certifications, and company requirements.

Interview & Assessment Management – Schedule assessments, mock interviews, technical rounds, mentoring sessions, and monitor participation.

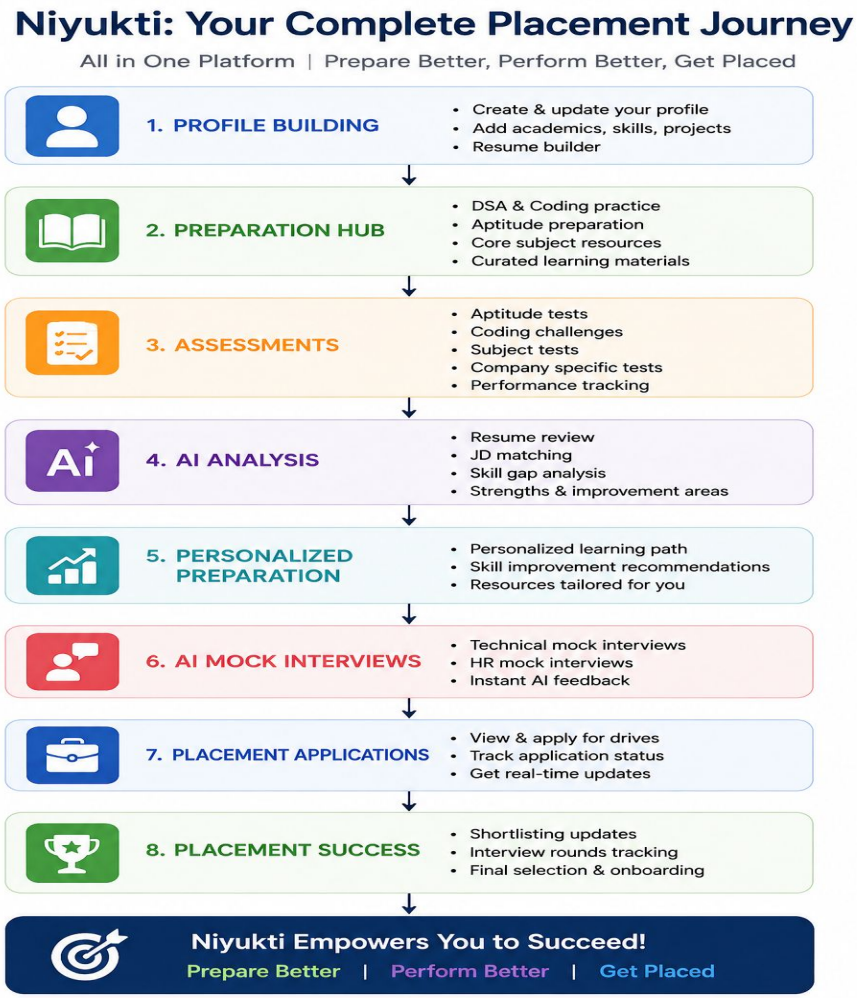
Analytics & Insights – Generate batch-wise and branch-wise reports, readiness analytics, placement statistics, and historical trends.

Placement Workflow: Company Added → Drive Created → Eligibility Defined → Applications Open → AI Shortlisting → Assessment → Technical Interview → HR Interview → Selection & Analytics



What Niyukti Offers to Students

Centralized Student Profile — Manage academic details, skills, projects, certifications, and resume in one place.
Personalized Placement Preparation — Access aptitude, coding, DSA, technical, and core-subject preparation.
Company-Specific Preparation — Prepare based on company requirements, roles, and placement patterns.
Assessments & Performance Tracking — Take assessments and monitor performance, progress, strengths, and areas for improvement.
AI Resume Analysis — Analyze resumes and identify areas for improvement.
Resume–Job Matching — Match student skills and profiles with relevant job roles and company requirements.
AI-Based Skill Gap Analysis — Identify missing skills and receive recommendations for improvement.
AI Mock Interviews — Practice technical and HR interviews with structured AI-generated feedback.
Placement Drive Access — View eligible companies, placement opportunities, and application details.
Application & Progress Tracking — Track applications, assessment status, interview stages, and placement progress.
Personalized Recommendations — Receive preparation suggestions based on performance, skills, and target roles.



AI Solutions for the Placement Cell

01 | JD INTELLIGENCE

AI analyses company JDs to identify eligibility, required skills, qualifications and role-specific requirements.

02 | SMART MATCHING & SHORTLISTING

Matches students with suitable roles using academic, skill, resume and assessment data — supporting the **Placement Cell** in shortlisting.

03 | STUDENT READINESS

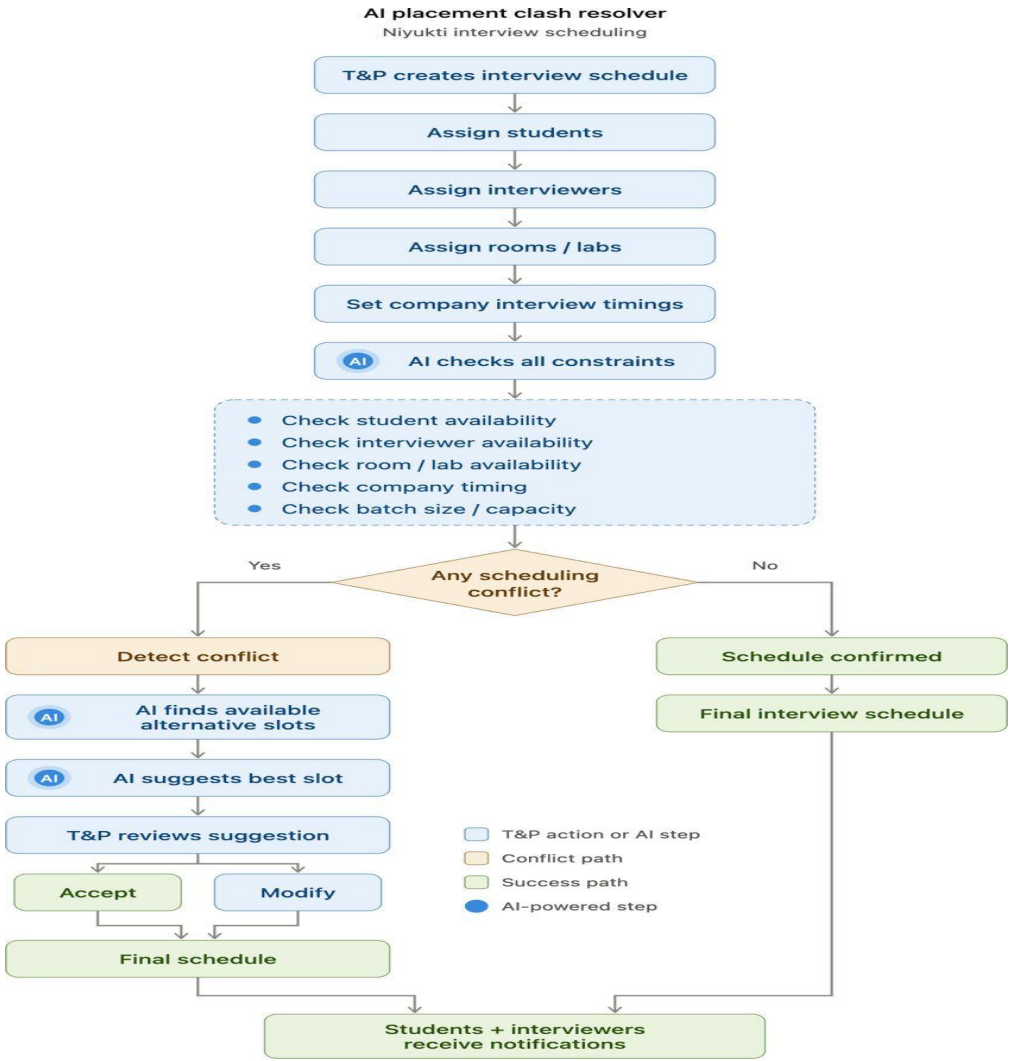
Analyses performance & skill gaps to recommend personalized preparation, resume improvements and AI-powered mock interviews.

04 | PLACEMENT CELL INTELLIGENCE

Converts placement data into insights on applications, assessments, selections, student readiness and drive-wise performance.

AI-DRIVEN PLACEMENT FLOW

Company JD → AI Analysis → Student–Job Matching → Student Preparation → Performance & Skill-Gap Analysis → T&P Shortlisting → Interview → Placement Insights



Additional Features

1 Attendance Monitoring System

A system to monitor and manage student attendance easily and maintain accurate attendance records.

2 Direct Company Requirements to Students

Company requirements and placement opportunities should be sent directly to eligible students instead of depending completely on Coordinators.

3 Placement Data Management

Maintain proper data of students placed from the LNCTU CSE department, including branch-wise, company-wise, and placement-wise records.

4 NBA Data Management

Properly track NBA-related data, faculty training, student training, and events conducted. This will help maintain accurate records and improve the NBA rating.

5 Timetable and Resource Clash Management

Avoid assigning two classes, events, or activities at the same time. The system should check whether the required time slot, classroom, lab, or building is available before assigning it.

6 Priority for Unplaced Students

Students who are already placed should be identified in the system, and preference should be given to eligible unplaced students for upcoming placement opportunities. This can help increase the number of uniquely placed students and improve the placement percentage.

7 Data Accuracy and Mismatch

Reduce problems such as copy-paste errors, duplicate data, data mismatch, and incorrect records by maintaining a centralized and validated database.

8 Communication Management

Create a proper communication system between the HOD, Principal, faculty members, and T&P coordinators so that information, tasks, and important updates can be shared and tracked easily.

9 AI Test Generation

The system allows faculty to upload a PDF containing the approved study material for the upcoming test. AI analyzes the content and generates relevant MCQS with options. The faculty reviews and approves the questions before they are sent to the students for assessment.

Challenges Faced

Personalized Roadmap Generation – Every student has different goals and skill levels, making it challenging to create relevant and customized learning paths.

Accurate AI Recommendations – Ensuring that AI suggestions are meaningful and aligned with student profiles, skills, and career aspirations.

Student Progress Tracking – Consolidating data from courses, projects, certifications, and assessments into a unified progress system.

AI-Based Shortlisting – Automatically matching students with company eligibility criteria while maintaining accuracy and fairness.

Scalability for T&P Departments – Managing and analyzing data for large numbers of students across multiple batches and branches.

Keeping Data Updated – Regularly updating learning resources, internship opportunities, and company requirements.

Student Engagement – Encouraging students to consistently update their profiles and actively use the platform.

Tech Stack

Mobile App: Kotlin • Jetpack Compose • MVVM • Retrofit • Coroutines/Flow • Hilt

Web Portal: React • TypeScript • Vite • Tailwind CSS • Recharts • Monaco Editor

Backend: Node.js • Express.js • TypeScript • REST APIs

Database & Storage: MongoDB • Mongoose • Atlas Vector Search • S3/MinIO

AI & Intelligence: OpenAI/Gemini • Resume Analysis • JD Matching • AI Recommendations

Security: JWT • Passport.js • RBAC • Zod

Assessments & Real-Time: Judge0 • Docker • Socket.IO • Redis • BullMQ

Communication & Reports: Nodemailer • Firebase Cloud Messaging • ExcelJS • Puppeteer

Testing & DevOps: Jest • Playwright • Docker • GitHub Actions • Sentry